

HDOC Day-30: Menu-Driven Array Operations in C

Question: Write a C program to perform menu-driven operations on a given array: (1) sort using Bubble Sort and perform Binary Search, (2) insertion and deletion at a valid position, (3) rotate the array to the right by k positions, and (4) exit.

Aim

To implement common array operations using a menu-driven C program and functions.

Algorithm

1. Read the size and elements of the array.
2. Display a menu repeatedly until the user selects Exit.
3. For option 1, sort the array using Bubble Sort, then use Binary Search for the required element.
4. For option 2, insert or delete an element after checking that the entered position is valid.
5. For option 3, rotate the array to the right by k positions using $(i + k) \% n$.
6. Stop when option 4 is selected.

C Program

```
#include <stdio.h>

#define MAX 100

void display(int a[], int n) {
    int i;
    printf("Array: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);
    printf("\n");
}

void bubbleSort(int a[], int n) {
    int i, j, temp;
    for (i = 0; i < n - 1; i++) {
        for (j = 0; j < n - i - 1; j++) {
            if (a[j] > a[j + 1]) {
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }
}

int binarySearch(int a[], int n, int key) {
    int low = 0, high = n - 1, mid;
    while (low <= high) {
        mid = (low + high) / 2;
        if (a[mid] == key)
            return mid;
        else if (a[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
}

void insertElement(int a[], int *n, int pos, int value) {
    int i;
    for (i = *n; i >= pos; i--)
        a[i] = a[i - 1];
    a[pos] = value;
    (*n)++;
}

void deleteElement(int a[], int *n, int pos) {
    int i;
    for (i = pos - 1; i < *n - 1; i++)
```



```

        scanf("%d", &value);
        insertElement(a, &n, pos, value);
        printf("After insertion:\n");
        display(a, n);
    }
    else if (subChoice == 2) {
        printf("Enter position (1 to %d): ", n);
        scanf("%d", &pos);

        if (pos < 1 || pos > n) {
            printf("Invalid position.\n");
            break;
        }

        deleteElement(a, &n, pos);
        printf("After deletion:\n");
        display(a, n);
    }
    else {
        printf("Invalid choice.\n");
    }
    break;

case 3:
    printf("Enter value of k: ");
    scanf("%d", &k);

    if (k < 0) {
        printf("Please enter a non-negative value of k.\n");
        break;
    }

    rotateRight(a, n, k);
    printf("Array after right rotation:\n");
    display(a, n);
    break;

case 4:
    printf("Program terminated.\n");
    break;

default:
    printf("Invalid choice. Try again.\n");
}

} while (choice != 4);

return 0;
}

```

Result

The program successfully performs Bubble Sort with Binary Search, insertion, deletion, and right rotation of the array through a menu-driven interface.